

Math 110 – Homework 13 Answers

Math 110 – Fall 2026

Homework 13

Complexification, real normal operators, and determinant

Suggested completion date: Friday, December 4

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let V be a finite-dimensional real vector space with basis $v_1, \dots, v_n$.

- (a) Prove that the same list is a basis of the complexification $V_{\mathbb{C}}$ over $\mathbb{C}$.
- (b) If $T \in \mathcal{L}(V)$, prove that the matrix of $T_{\mathbb{C}}$ with respect to this basis is the same matrix as that of T .

Problem 2. Let $T \in \mathcal{L}(V)$ for a finite-dimensional real vector space V . Prove that T and its complexification $T_{\mathbb{C}}$ have the same minimal polynomial.

Problem 3. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

act on $\mathbb{R}^2$, and consider its complexification.

- (a) Find the two complex eigenvalues and corresponding eigenvectors of $J_{\mathbb{C}}$.
- (b) Explain how the two eigenvectors are related by complex conjugation.
- (c) Explain how this example gives the real 2×2 rotation block associated with a conjugate pair of complex eigenvalues.

Problem 4. Let

$$A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

act on $\mathbb{R}^2$.

- (a) Prove directly that A is normal.
- (b) Find the eigenvalues of $A_{\mathbb{C}}$.
- (c) Prove that A is an isometry if and only if $a^2 + b^2 = 1$.

Problem 5. Let T be an operator on a two-dimensional vector space with basis e_1, e_2 , and suppose

$$\mathcal{M}(T) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Using the alternating-form definition of determinant, recover the formula

$$\det T = ad - bc.$$

Problem 6 (optional). Let T be an operator on a finite-dimensional complex vector space. Using a Jordan basis and the formula for the determinant of an upper-triangular matrix, prove

$$\det(zI - T) = \chi_T(z),$$

where χ_T is the characteristic polynomial defined earlier using algebraic multiplicities.

Problem 7 (optional). Let

$$F_0 = 0, \quad F_1 = 1, \quad F_{n+1} = F_n + F_{n-1},$$

and let

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Using

$$A^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix} \quad (n \geq 1)$$

and multiplicativity of determinant, prove Cassini's identity

$$F_{n-1}F_{n+1} - F_n^2 = (-1)^n.$$

Problem 8 (optional). Let T have singular values $s_1, \dots, s_n$. Using the singular value decomposition, prove

$$|\det T| = s_1 \cdots s_n.$$

Solutions

Problem 1

Solution notes

Problem 2

Solution notes

Problem 3

Solution notes

Problem 4

Solution notes

Problem 5**Solution notes**

Problem 6**Solution notes**

Problem 7**Solution notes**

Problem 8**Solution notes**
